07/01/2026

Code-A Phase-1

Aakash
Medical | IIT-JEE | Foundations

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**MM : 720**

NCERT Booster Program for NEET-2026_NBTS-03A

**Time : 180 Min.**

**Topics Covered:**
**Physics:** Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory
**Chemistry:** Equilibrium, Redox Reactions, Some p Block Elements (Group-13 and 14), Organic Chemistry: Some Basic Principles and Techniques, Hydrocarbons
**Botany:** Anatomy of Flowering Plants, Plant Kingdom, Photosynthesis in Higher Plants, Respiration in Plants
**Zoology:** Locomotion and Movement, Neural Control and Coordination, Chemical Coordination and Integration

**General Instructions :**
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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## PHYSICS

1. A planet revolving around the sun sweeps area $A_1$ in 2 days, $A_2$ in 4 days and $A_3$ in 8 days. The ratio $A_1 : A_2 : A_3$ is

   (1) 1 : 2 : 4

   (2) 1 : 4 : 8

   (3) 2 : 5 : 6

   (4) 4 : 8 : 1

2. The escape velocity of an object projected from surface of a given planet is independent of

   (1) Radius of planet

   (2) Mass of planet

   (3) Mass of the object

   (4) Both (2) and (3)

3. **Assertion (A):** A planet moves faster, when it is closer to the sun in its orbit and vice-versa.
   **Reason (R):** Areal velocity of the planet in an orbit is constant.

   (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

   (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

   (3) Assertion is true statement but Reason is false

   (4) Both Assertion and Reason are false statements

4. A body weighs 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is :

   (1) 32 N

   (2) 36 N

   (3) 16 N

   (4) 27 N

5. **Assertion (A) :** In stress-strain graph, brittle materials have large region beyond elastic limit.
**Reason (R) :** Beyond yield point, body does not regain its original size.
In the light of the above statements, choose the correct answer from the options given below :

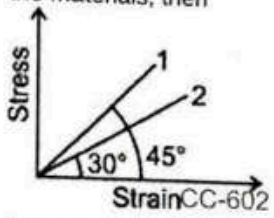
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

6. For two materials, the stress versus strain graphs is drawn as shown in figure. If $Y_1$ and $Y_2$ are the Young's modulus of the materials, then




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(1) $Y_1 = 2Y_2$

(2) $Y_1 = \frac{\sqrt{3}}{2}Y_2$

(3) $Y_1 = \sqrt{3}Y_2$

(4) $Y_1 = \frac{Y_2}{\sqrt{3}}$

7. The Bulk modulus for the perfectly rigid body is

(1) 1

(2) Zero

(3) Infinite

(4) $\frac{1}{2}$

8. Factors affecting modulus of elasticity of any material is/are

(1) Temperature

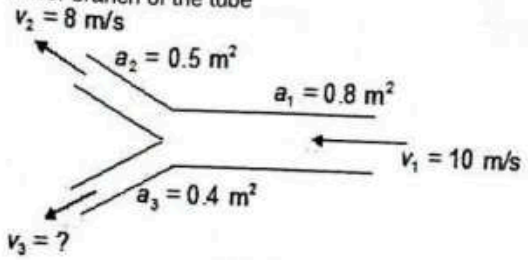
(2) Nature of material

(3) Impurity

(4) All of the above

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9. Calculate the velocity of flow of the ideal liquid ($v_3$) in the lower branch of the tube




(1) 4 m/s

(2) 6 m/s

(3) 8 m/s

(4) 10 m/s

10. Sixty four spherical rain drops of equal size are falling vertically through air with a terminal speed of 2.5 m/s. If these drops coalesce to form a big spherical drop, then terminal velocity of big drop is

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(1) 10 m/s

(2) 40 m/s

(3) 20 m/s

(4) 30 m/s

11. Temperature of a gas is increased from 127°C to 527°C. The rms speed of gas molecules will increase to a factor of

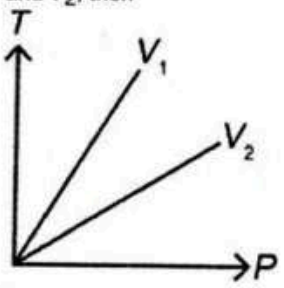
(1) 2

(2) 4

(3) $\sqrt{2}$

(4) $\sqrt{3}$

12. The figure shows temperature ($T$) versus pressure ($P$) graphs for certain amount of ideal gas at two volumes $V_1$ and $V_2$, then




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(1) $V_1 = V_2$

(2) $V_1 < V_2$

(3) $V_1 > V_2$

(4) Insufficient information

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**13.** A gaseous mixture consists of 16 g of $O_2$ and 16 g of He at temperature $T$. Neglecting the vibrational modes, total internal energy of the system is

(1) 24 $RT$

(2) 17 $RT$

(3) 29 $RT$

(4) $\frac{29}{4}RT$

**14.** The mean free path ($\lambda$) for a gas molecule depends upon number density ($n$) as

(1) $\lambda \propto n$

(2) $\lambda \propto \frac{1}{n}$

(3) $\lambda \propto \frac{1}{n^2}$

(4) $\lambda \propto \frac{1}{\sqrt{n}}$

**15.** If the rms speed of helium gas molecules at temperature $T_0$ and pressure $P_0$ is 898 m/s, then the rms speed of helium molecules at temperature $T_0$ and pressure $2P_0$ will be

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(1) 450 m/s

(2) 898 m/s

(3) 980 m/s

(4) 1800 m/s

**16.** In a thermodynamic system, internal energy decreases by 300 J when it is doing 350 J of work. What net heat is taken by system in the process?

(1) 150 J

(2) 650 J

(3) 100 J

(4) 50 J

**17.** The source and sink temperature of a Carnot engine are 327°C and 27°C respectively. The efficiency of the engine is

(1) 60%

(2) 55%

(3) 70%

(4) 50%

**18.** In a container, pressure of a gas is measured as 100 kPa. If one-fourth of the gas is removed from the container keeping temperature constant, then pressure of the remaining gas will be

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(1) 25 kPa

(2) 50 kPa

(3) 75 kPa

(4) 100 kPa

**19.** In an adiabatic process, the volume and temperature of a monoatomic gas are related as $V \propto T^C$, where $c$ is equal to

(1) $-\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) $-\frac{3}{2}$

**20.** An ideal gas is taken through a process given by $P = KV^2$, where $P$ is pressure and $V$ is volume. The molar heat capacity of the gas for the process is [$K$ is a positive constant]

(1) $C_v + \frac{R}{2}$

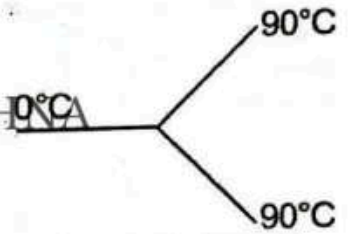
(2) $C_p - \frac{2R}{3}$

(3) $C_p + \frac{R}{2}$

(4) $C_p + \frac{R}{3}$

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**21.** Three rods made of same material and having same cross-section have been joined as shown in figure. Each rod is of same length. The temperature at the junction of three rods is




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(1) 60°C

(2) 45°C

(3) 90°C

(4) 30°C

**22.** If $\lambda_m$ denotes the wavelength at which the radiative emission from a black body at a temperature $T$ K is maximum, then

(1) $\lambda_m \propto T^4$

(2) $\lambda_m \propto T^0$

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(3) $\lambda_m \propto T^{-1}$

(4) $\lambda_m \propto T$

**23.** A satellite of mass $m$ is orbiting the earth at an altitude of $\frac{R}{2}$, where $R$ is radius of earth. If extra energy equal to $\frac{mgR}{12}$ is given to satellite, then increase in its altitude is

(1) $R$

(2) $\frac{R}{2}$

(3) $2R$

(4) $\frac{R}{3}$

**24.** An ideal gas at 127°C is compressed adiabatically to $\frac{8}{27}$ of its original volume. The percentage rise in temperature is $\left(\gamma = \frac{5}{3}\right)$.

(1) 50%

(2) 75%

(3) 100%

(4) 125%

**25.** In a U-tube, $h$ height of water column is balanced by $h_0$ height of an oil column. If relative density of oil is $\frac{6}{5}$, then $\frac{h}{h_0}$ is

(1) $\frac{6}{5}$

(2) $\frac{5}{6}$

(3) $\frac{5}{11}$

(4) $\frac{6}{11}$

**26.** If $X$ g of water at 50°C is mixed with 10 g of ice at 0°C, then minimum value of $X$ for complete melting of ice is

(1) 20

(2) 40

(3) 16

(4) 100

**27.** Temperature at which average kinetic energy of helium atom is $6.21 \times 10^{-21}$ J, is ($K_B = 1.38 \times 10^{-23}$ J K$^{-1}$).

(1) 20°C

(2) 47°C

(3) 27°C

(4) 32°C

**28.** Water rises to a height of 10 cm in a capillary tube of radius $r$. If radius of tube made of same material is increased to $3r$, then the height to which water will rise in the tube is

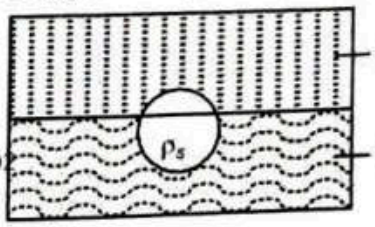
(1) 30 cm

(2) 10 cm

(3) $\frac{10}{3}$ cm

(4) $\frac{10}{\sqrt{3}}$ cm

**29.** A vessel contains two types of liquids $A$ and $B$ as shown in figure. A homogeneous sphere of density $\rho_S$ floats with one-third of its volume immersed in liquid $A$ and the rest two-third of its volume in liquid $B$. If density of liquid $A$ is $\rho_A$ and liquid $B$ is $\rho_B$, the density of the sphere is ($\rho_B > \rho_A$).




(1) $\frac{\rho_A + \rho_B}{3}$

(2) $\frac{\rho_A + \rho_B}{2}$

(3) $\frac{2\rho_A + \rho_B}{3}$

(4) $\frac{\rho_A + 2\rho_B}{3}$

**30.** Young's modulus of a rod is $5 \times 10^{10}$ N/m$^2$. The rod is deformed by a force. If potential energy stored per unit volume of the rod is 4000 J/m$^3$, then strain produced in the rod is

(1) $4 \times 10^{-3}$

(2) $2 \times 10^{-4}$

(3) $4 \times 10^{-4}$

(4) $3 \times 10^{-3}$

**31.** Two rods of equal length and area of cross-section are kept parallel and their ends are maintained between temperature 20°C and 80°C. The ratio of the effective thermal conductivity to that of the first rod is

$\left[\text{the ratio } \left(\frac{K_1}{K_2}\right) = \frac{3}{4}\right]$

(1) 7:4

(2) 7:6

(3) 4:7

(4) 7:8

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**32.** An ideal gas at 27°C temperature and 10 atmospheric pressure is allowed to expand till its final pressure becomes equal to atmospheric pressure. If the volume becomes 10 times of its initial volume, then change in its temperature will be

(1) 0°C

(2) 10°C

(3) 15°C

(4) 20°C

**33.** A container filled with water upto height of 50 cm is tied to a rope which passes over a frictionless light pulley (like an Atwood machine). The other end of the rope is tied to a mass which is one-third of the mass of the water filled container. The gauge pressure at the bottom of the container is ($g$ = 10 m/s$^2$).

(1) 2 kPa

(2) 2.5 kPa

(3) 1.5 kPa

(4) 5 kPa



**34.** A liquid cool from 61°C to 59°C in two minutes. It cools from 31°C to 29°C in five minutes. If Newton's law of cooling is obeyed by the liquid, then temperature of surroundings will be

(1) 20°C

(2) 15°C

(3) 10°C

(4) 22°C

**35.** Power radiated by a black body is $P$ and it radiates maximum energy at wavelength $\lambda_0$. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{2}{3}\lambda_0$, the power radiated by it becomes $XP$. The value of $4X$ is

(1) $\frac{81}{16}$

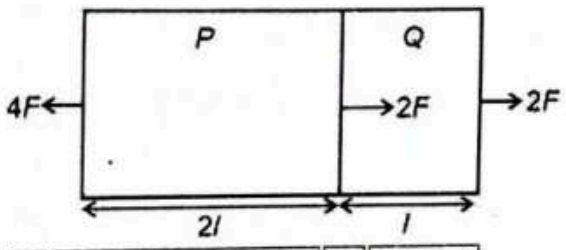
(2) $\frac{23}{81}$

(3) $\frac{81}{4}$

(4) $\frac{256}{81}$



**36.** A bar is subjected to axial forces as shown. If $Y$ is the modulus of elasticity of the bar and $A$ is its cross-section area, then match column-I with column-II.




| | Column-I | | Column-II |
|---|---|---|---|
| (A) | Elongation in P | (P) | $\frac{2Fl}{YA}$ |
| (B) | Elongation in Q | (Q) | $\frac{10Fl}{YA}$ |
| (C) | Total elongation of the bar | (R) | $\frac{8Fl}{YA}$ |

(1) A(R), B(Q), C(P)

(2) A(Q), B(P), C(R)

(3) A(R), B(P), C(Q)

(4) A(P), B(Q), C(R)



**37.** Two planets of masses $M_1$ and $M_2$ have satellites of masses $m_1$ and $m_2$ respectively, revolving around them at the same orbital radius. The period of satellite $m_1$ is thrice as that of the second satellite $m_2$. Select the correct relations among the following

(1) $9m_1 = m_2$

(2) $9m_2 = m_1$

(3) $9M_1 = M_2$

(4) $9M_2 = M_1$

**38.** A faulty thermometer reads 10°C in melting ice and 106°C in steam. The temperature in °F, when this faulty thermometer reads 50°C is

(1) 110°F

(2) 107°F

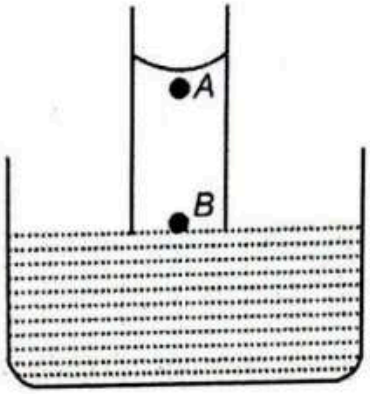
(3) 104°F

(4) 116°F



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**39.** A capillary tube is dipped in water as shown. If $P$ is the atmospheric pressure, $P_A$ the pressure at $A$ and $P_B$ the pressure at $B$, then choose the correct option.




(1) $P_A = P_B = P$

(2) $P_A > P_B > P$

(3) $P > P_A$ , $P = P_B$

(4) $P_A = P$, $P > P_B$

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**40.** A pendulum clock with a pendulum made of invar material having thermal coefficient $\alpha = 0.7 \times 10^{-6} {}^{\circ}C^{-1}$ has a period of 0.6 s and is accurate at 25°C. If clock is used at a place where average temperature is 30°C, what correction is necessary at the end of a month (30 days) to the time given by the clock?

(1) 9.10 s

(2) 3.85 s

(3) 8.60 s

(4) 4.53 s

**41.** A body having surface area 5 $cm^2$ and a temperature of 727°C radiates 500 joule of energy each minute. Emissivity of the body is $\left(\sigma = 5.67 \times 10^{-8} \frac{W}{m^2 k^4}\right)$

(1) 0.15

(2) 0.25

(3) 0.35

(4) 0.20

**42.** The maximum load a wire can withstand without breaking is $F$. If the length is doubled, then the maximum load the new wire of the same material can withstand is

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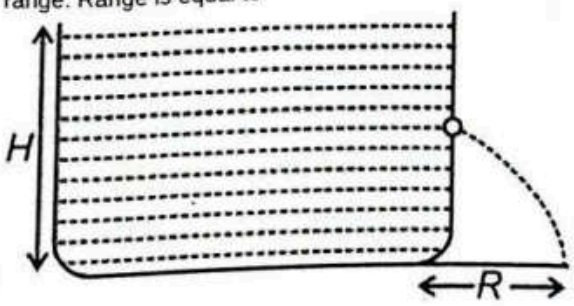
(1) $F$

(2) $2F$

(3) $\frac{F}{2}$

(4) $4F$

**43.** A container is filled with water upto height $H$ as shown in figure. There is a hole in the container such that water coming out from it has range half of the maximum possible range. Range is equal to




(1) $R = H$

(2) $R = \frac{H}{2}$

(3) $R = \frac{H}{3}$

(4) $R = \frac{H}{4}$

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**44.** Consider the given statements.

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**Statement-A:** Temperature of a body is related to its internal energy, not to the kinetic energy of motion of its centre of mass.

**Statement-B:** Heat capacity of a gas, in general, is independent of the process the system goes through when heat is supplied.

**Statement-C:** Equilibrium states of a thermodynamic system are described by state variables.

Which of the following statement(s) is/are correct.

(1) A and B only

(2) B and C only

(3) A and C only

(4) All A, B and C

**45.** Work done by an ideal gas in an adiabatic change of state from $(P_1, V_1, T_1)$ to $(P_2, V_2, T_2)$ is

(1) $W = \frac{nR(T_2 - T_1)}{\gamma - 1}$

(2) $W = \frac{nR(T_1 - T_2)}{\gamma + 1}$

(3) $W = \frac{nR(T_1 + T_2)}{\gamma - 1}$

(4) $W = \frac{nR(T_1 - T_2)}{\gamma - 1}$

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**CHEMISTRY**

**46.** Heterogeneous equilibrium among the following is

(1) $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$

(2) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(3) $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$

(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

**47.** If $K_b$ of $NH_3$ is $2 \times 10^{-5}$ at 25°C then $K_a$ for $\overset{+}{N}H_4$ ion is

(1) $2 \times 10^{10}$

(2) $5 \times 10^{-10}$

(3) $2 \times 10^{-19}$

(4) $5 \times 10^{-19}$

**48.** For the reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, the forward reaction at constant temperature is favoured by

(1) Introducing $PCl_3(g)$ at constant volume

(2) Introducing inert gas at constant volume

(3) Introducing $PCl_5(g)$ at constant volume

(4) Introducing $Cl_2(g)$ at constant volume

**49.** Match List-I with List-II.

| | List-I (Salt) | | List-II (Ksp) |
|---|---|---|---|
| (a) | AgBr | (i) | $27s^4$ |
| (b) | $Hg_2Cl_2$ | (ii) | $4s^3$ |
| (c) | $Al(OH)_3$ | (iii) | $s^2$ |
| (d) | $Ca_3(PO_4)_2$ | (iv) | $108s^5$ |

Choose the **correct** answer form the options given below

(1) a(iv), b(ii), c(iii), d(i)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(i), b(iii), c(iv), d(ii)

**50.** The percent ionisation of 0.01 M acetic acid in 0.1M HCl will be ($K_a$ of acetic acid is $1.8 \times 10^{-5}$)

(1) 0.18%

(2) 0.009%

(3) 0.018%

(4) 0.09%

**51.** If equal volumes of two solutions of pH = 2 and pH = 3 are mixed, then the pH of resulting solution will be (log 5.5. = 0.74)

(1) 0.74

(2) 2.74

(3) 3.26

(4) 2.26

**52.** The pair of compounds which can make basic buffer is

(1) $NH_4OH$ & $NH_4Cl$

(2) $CH_3COOH$ & $CH_3COONa$

(3) $H_2CO_3$ & $Na_2CO_3$

(4) $HClO_4$ & $NaClO_4$

**53.** For dissociation of $NH_4HS(s)$ into $H_2S(g)$ and $NH_3(g)$,

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$$

The total pressure at equilibrium is 20 atm. The $K_c$ for this equilibrium will be

(1) $10(RT)^2$

(2) $100(RT)^{-2}$

(3) 10RT

(4) 100 RT

**54.** Consider the thermal dissociation of solid $A_2B_3$

$A_2B_3(s) \rightleftharpoons 2\,AB(g) + B(g)$

If total pressure at equilibrium is 3 atm then $K_p$ of the reaction will be

(1) 1

(2) 3

(3) 9

(4) 4

**55.** The pH of 0.1 M ammonium acetate solution at 298 K is ($pK_a$ of $CH_3COOH$ = 4.7, $pK_b$ of $NH_4OH$ = 4.7)

(1) 6.9

(2) 7

(3) 7.1

(4) 13.9

**56.** If 1 L aqueous solution of 0.05 M $H_2SO_4$ is diluted to 1000 L then pH of the final solution is

(1) 2

(2) 3

(3) 4

(4) 3.5

**57.** The molar solubility of $MgF_2$ ($K_{sp} = 6.5 \times 10^{-9}$) in 0.1 M solution of NaF will be

(1) $6.5 \times 10^{-5}$

(2) $6.5 \times 10^{-7}$

(3) $6.5 \times 10^{-11}$

(4) $6.5 \times 10^{-8}$

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**58.** Standard reduction potentials of the half reactions are given below

$Ag^+ + e^- \rightarrow Ag \quad E^0 = 0.8\ V$

$Cr^{3+} + 3e^- \rightarrow Cr \quad E^0 = -0.74\ V$

$Zn^{2+} + 2e^- \rightarrow Zn \quad E^0 = -0.76\ V$

The strongest oxidising and reducing species respectively are

(1) $Ag^+$, $Zn^{2+}$

(2) Ag, Cr

(3) Ag, $Cr^{3+}$

(4) $Ag^+$, Zn

**59.** Which among the following species cannot act as a reducing agent?

(1) $SO_4^{2-}$

(2) $SO_3^{2-}$

(3) $S_8$

(4) $H_2S$

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**60.** Balance the given redox reaction in acidic medium and match the reactants/products given in column I with their coefficient in column II.

$S_2O_3^{2-} + Br_2 + H_2O \rightarrow SO_4^{2-} + Br^- + H^+$

| Column I | Column II |
|---|---|
| a. $S_2O_3^{2-}$ | (i) 10 |
| b. $Br_2$ | (ii) 8 |
| c. $H^+$ | (iii) 4 |
| d. $Br^-$ | (iv) 1 |

Choose the **correct** option

(1) a(iv), b(iii), c(ii), d(i)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(i), d(ii)

**61.** If oxidation number of X, Y, Z in a compound are –3, +5 and –1 respectively then possible formula of compound will be

(1) $X_2(YZ)_2$

(2) $X_3(YZ_2)_2$

(3) $X(Y_2Z)_2$

(4) $X(YZ_4)_3$

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**62.** Match list I with list II

| | List I (Species) | | List II (Oxidation state of central atom) |
|---|---|---|---|
| a. | $C_3O_2$ | (i) | +7 |
| b. | $Br_3O_8$ | (ii) | +5 |
| c. | $HClO_4$ | (iii) | 0 |
| d. | $H_3PO_4$ | (iv) | +4 |

Choose the correct option

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

**63.** $E^\circ_{cell}$ of the Daniel cell is

(Given $E^\circ_{Zn^{2+}/Zn} = -0.76\ V,\ E^\circ_{Cu^{2+}/Cu} = 0.34\ V$)

(1) 0.42 V

(2) –0.42 V

(3) 1.1 V

(4) –1.1 V

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**64.** Match the following and choose the correct option.

| Column I | Column II |
|---|---|
| a. Inorganic benzene | (i) $Na_2B_4O_7.10H_2O$ |
| b. Diborane | (ii) $H_3BO_3$ |
| c. Borax | (iii) $B_3N_3H_6$ |
| d. Orthoboric acid | (iv) $B_2H_6$ |

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(iv), c(ii), d(i)

**65.** Given below are two statements one is assertion (A) other is reason (R).

**Assertion :** +2 oxidation state of Pb is more stable its than +4 oxidation state.

**Reason :** Pb shows inert pair effect.

In the light of above statements, choose the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

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(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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**66.** Given below are two statements, one is labelled as Assertion (A) other is labelled as Reason (R).

Assertion(A) : Boron cannot form $[BF_6]^{3-}$.

Reason(R) : Maximum covalency of B is 4 due to the absence of vacant d-orbitals.

In the light of above statements, choose the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**67.** Consider the following statements about allotropes of carbon

(i) Diamond is thermodynamically most stable allotrope of carbon

(ii) Graphite conducts electricity

(iii) Buckminsterfullerene does not contain 'dangling' bonds

The correct statements are

(1) (i), (ii) and (iii)

(2) (ii) and (iii) only

(3) (i) and (ii) only

(4) (i) and (iii) only

**68.** Total number of six- membered rings and five-membered rings present in fullerene (C-60) respectively are

(1) 12 and 20

(2) 16 and 20

(3) 20 and 15

(4) 20 and 12

**69.** Which of the following oxides is an acidic in nature?

(1) $SiO_2$

(2) $In_2O_3$

(3) $Ga_2O_3$

(4) CO

**70.** Water gas among the following is

(1) $CO_2 + H_2$

(2) $CO + H_2$

(3) $CO + N_2$

(4) $CO_2 + N_2$

**71.** Match the list I with list II.

| | List I (Compounds) | | List II (Type of Isomerism) |
|---|---|---|---|
| a. | Butan-1-ol and Butan-2-ol | (i) | Metamerism |
| b. | n-pentane & neopentane | (ii) | Functional group isomerism |
| c. | Methoxy propane & Ethoxyethane | (iii) | Chain isomerism |
| d. | Propanone and Propanal | (iv) | Position isomerism |

Choose the most appropriate match from the options given below:

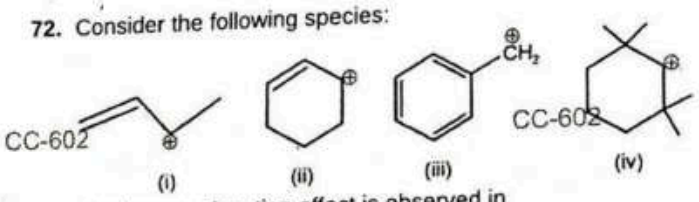
(1) a – (iii), b–(iv), c–(i), d–(ii)

(2) a – (iii), b–(ii), c–(i), d–(iv)

(3) a – (iv), b–(iii), c–(ii), d–(i)

(4) a –(iv), b–(iii), c–(i), d–(ii)

**72.** Consider the following species:




(i) (ii) (iii) (iv)

The hyperconjugation effect is observed in

(1) (i) and (iii) only

(2) (i) and (ii) only

(3) (ii), (iii) and (iv) only

(4) (i), (ii), (iii) and (iv)

**73.** In sulphur estimation, 1.5 g of an organic compound gave 0.5825 of barium sulphate. The percentage of sulphur in the organic compound is (molar mass of $BaSO_4$ = 233 g/mol)

(1) 5.33%

(2) 10.66%

(3) 12.8%

(4) 24.2%

**74.** Consider the following statements

**Statement I:** Ammonium phosphomolybdate is yellow in colour.

**Statement II:** The colour of the complex $[Fe(CN)_5NOS]^{4-}$ is black.

In the light of above statements, choose the correct option.

(1) Both the statement I and statement II are correct

(2) Both the statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

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**75.** During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.4 g of compound was neutralized by 10 mL of 0.2 M $H_2SO_4$ solution. The percentage of nitrogen in the compound is

(1) 14%

(2) 18%

(3) 28%

(4) 35%

**76.** The correct order of stability of the given carbocations is

(a) $(CH_3)_3C^+$

(b) $CH_3-\overset{+}{C}H_2$

(c) $(CH_3)_2\overset{\oplus}{C}H$

(1) (a) > (b) > (c)

(2) (b) > (a) > (c)

(3) (a) > (c) > (b)

(4) (b) > (c) > (a)

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**77.** Given below are the two statements one is labelled as Assertion (A) and other is labelled as Reason (R).

**Assertion (A):** Melting point of trans-but-2-ene is higher than that of cis-but-2-ene

**Reason (R):** Dipole moment of trans-but-2-ene is higher than that of cis-but-2-ene

In the light of above statements, choose the **correct** options from the answer given below.

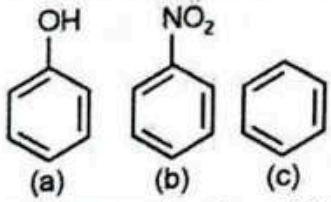
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

**78.** Consider the following species:




The correct order of reactivity of the given compounds towards electrophilic substitution reaction is

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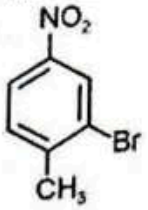
(1) (a) > (b) > (c)

(2) (b) > (c) > (a)

(3) (a) > (c) > (b)

(4) (c) > (a) > (b)

**79.** The IUPAC name of the given organic compound is




(1) 1-Bromo-4-nitro-toluene

(2) 1-Bromo-2-methyl-5-nitrobenzene

(3) 2-Bromo-1-methyl-4-nitrobenzene

(4) 4-methyl-5-bromo-1-nitrobenzene

**80.** Match list I with list II.

| | List I (Mixture) | | List II (Separation Technique) |
|---|---|---|---|
| a. | Aniline from aniline-water mixture | (i) | Fractional distillation |
| b. | Glycerol from spent-lye | (ii) | Distillation |
| c. | Chloroform and Aniline | (iii) | Distillation under reduced pressure |
| d. | Different fractions of crude oil | (iv) | Steam distillation |

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Choose the most appropriate match from the options given below:

(1) a–(iv), b–(iii), c–(i), d–(ii)

(2) a–(iv), b–(iii), c–(ii), d–(i)

(3) a–(iii), b–(iv), c–(i), d–(ii)

(4) a–(iii), b–(i), c–(ii), d–(iv)

**81.** Choose the incorrect match:

(1) Tropone – Non benzenoid aromatic compound

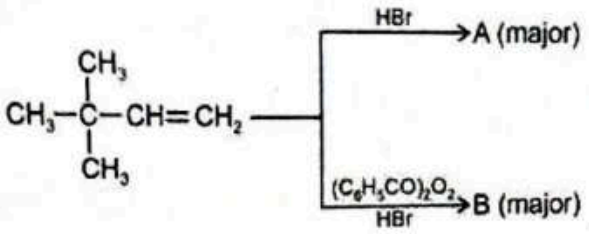
(2) Naphthalene – Benzenoid aromatic compound

(3) 1, 3-Cyclobutadiene – Antiaromatic compound

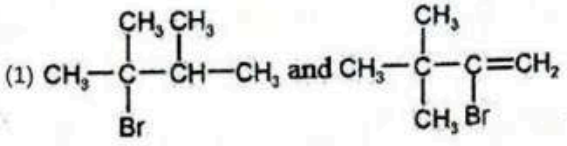
(4) Cyclooctatetraene – Aromatic compound

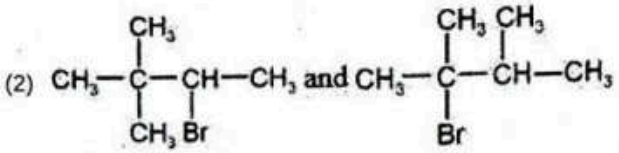
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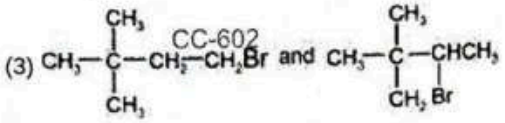
**82.** Consider the given reactions:

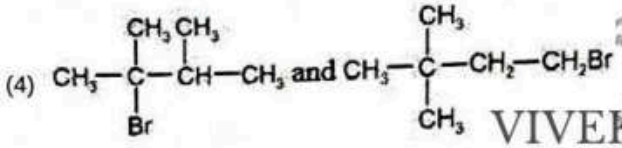



Major products (A) and (B) respectively are

(1) 

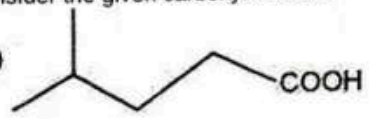
(2) 

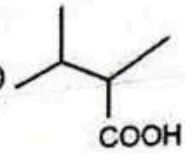
(3) 

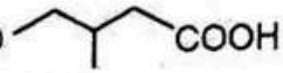
(4) 

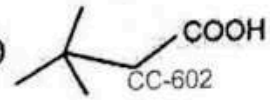
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**83.** Consider the given carboxylic acids:

(a) 

(b) 

(c) 

(d) 

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Sodium salt of which of the above carboxylic acids will undergo decarboxylation with sodalime to form isopentane as a major product?

(1) a, b and c only

(2) b and c only

(3) b, c and d only

(4) a, b, c and d

**84.** Which one of the following reactions does not give benzene as a major product?

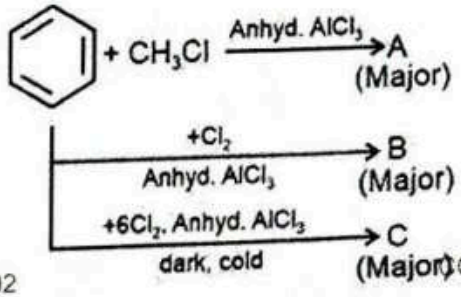
(1) Heating of n-hexane at 773 K and 10-20 atmospheric pressure in the presence of $V_2O_5$ supported over alumina.

(2) Heating of n-hexane in the presence of anhydrous aluminium chloride and hydrogen chloride gas

(3) Heating of phenol with zinc dust

(4) Cyclic polymerisation of ethyne using red hot iron tube at 873 K temperature

**85.** Consider the given reactions:




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A, B and C respectively are

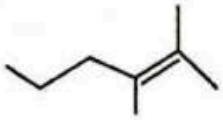
(1) Toluene, chlorobenzene, benzene hexachloride

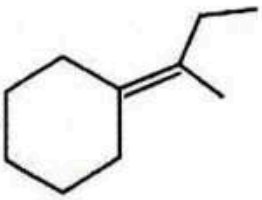
(2) Toluene, chlorobenzene, hexachlorobenzene

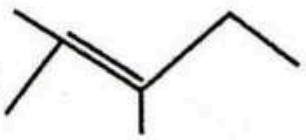
(3) Chlorobenzene, hexachlorobenzene, benzene hexachloride

(4) Benzene hexachloride, chlorobenzene, hexachlorobenzene

**86.** Which of the following compounds will not give acetone as one of the products on reductive ozonolysis?

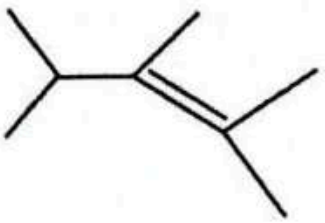
(1) 

(2) 

(3) 

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(4) 

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87. Choose the incorrect statement

(1) In ethyne, both the hydrogen atoms are acidic

(2) Electronegativity of carbon atom with sp hybridisation is more as compared to that with $sp^2$ hybridisation

(3) Ethyne is more acidic than propyne

(4) But-2-yne is more acidic than propyne

88. The major product 'A' of the given reaction is

$$(CH_3)_2C = CH_2 \xrightarrow[\Delta]{KMnO_4/H^+} A + CO_2 + H_2O$$

(1) Propan-2-one

(2) Propanal

(3) Propan-1,2-diol

(4) 2-Methylpropan-2-ol

89. Which of the given compounds is not a hydrocarbon?

(1) Anisole

(2) Xylene

(3) Decane

(4) Toluene

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90. Which of the given carboxylic acids is most acidic?

(1) $Cl_3CCOOH$

(2) $ClCH_2COOH$

(3) $Cl_2CHCOOH$

(4) $(CH_3)_3CCOOH$

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## BOTANY

91. The epidermal tissue system forms the outer-most covering of the plant body and it

(1) Bears number of hairs and chloroplast containing structures in shoot system

(2) Is always covered by thick and water impermeable substance called suberin

(3) Is made up of parenchymatous tissue with large intercellular spaces

(4) Is usually multilayered and comprised of cells that lack vacuoles

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92. Read the following statements.

(a) The dicots possess the ability to form secondary xylem and phloem tissues.

(b) Secondary tissues are produced by the activity of vascular cambium in dicots.

(c) Initiation of lateral roots and vascular cambium during the secondary growth is facilitated by the pericycle.

(d) Part of ground tissue in the stele which is located between the phloem and the endodermis is sclerenchymatous in higher plants.

Which of these is true?

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(1) (a) and (b) are true and (a) is the correct reason for (b)

(2) (b) and (c) are true and (c) is the correct reason for (b)

(3) (a) and (b) are true and (b) is the correct reason for (a)

(4) (c) and (d) are true and (d) is the correct reason for (c)

**93.** A biologist studied the anatomy of flowering plant parts labelled as A, B and C. Given below are the observations:

| Plant Part | Observation |
|---|---|
| (A) | Central portion is large and well developed and occupied by the component of ground tissue system. |
| (B) | The transverse and radial walls of the single layered barrel-shaped cells are thickened with water impermeable substance. |
| (C) | Vascular bundles are surrounded by a layer of thick-walled chloroplast containing cells with no intercellular spaces. |

From the observation, what can be the most probable inferences that might be drawn?
(i) The plant part 'A' may be stem or root of monocot
(ii) The plant part 'A' and 'B' can be of same plant
(iii) The plant part 'C' can be isobilateral leaf
(iv) The plant part 'A', 'B' and 'C' respectively can be the parts of stem, root and leaf of dicot

(1) Only (i)
(2) Only (i), (ii) and (iv)
(3) Only (i) and (ii)
(4) Only (ii), (iii) and (iv)

**94.** During redox reaction in glycolysis, which molecule acts as the oxidizing agent?

(1) $NAD^+$
(2) $NADH_2$
(3) ATP
(4) $FADH_2$

**95.** Identify the feature that is **not** true for dicot stem.

(1) Cortex consists of hypodermis, cortical layers and endodermis.
(2) A large number of vascular bundles are arranged in a ring.
(3) Vascular bundles are conjoint and open and with metaxylem towards periphery.
(4) Conjuctive tissue that lies between the xylem and the phloem have lignin in their cell wall.

**96.** Select the **correctly** matched pair.

(1) Epiblema – Forms unicellular elongations that helps to absorb water and minerals from the soil.
(2) Cuticle – Water impermeable substance that forms a thick layer to cover the root, stem and leaves of higher plants.
(3) Palisade parenchyma – Constituent cells are placed beneath the abaxial epidermis and are arranged vertically and parallel to each other.
(4) Guard cell – Specialised cell that surrounds the non-photosynthetic subsidiary cells.

**97.** The transverse section of a plant part showed polyarch, radial and exarch xylem. This plant part can be

(1) Dicot stem
(2) Monocot stem
(3) Monocot root
(4) Dicot root

**98.** In monocot stem, the hypodermis is made up of

(1) Parenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Chlorenchyma

**99.** In between the vascular bundles of dicot stem, there are a few layers of radially placed parenchymatous cells, which constitute

(1) Bundle sheath
(2) Medullary rays
(3) Cambium
(4) Pericycle

**100.** Which of the given features is **true** for dicot leaf?

(1) The adaxial epidermis never lack stomata
(2) The oval or round parenchyma cells having large air cavities between them are situated above the adaxial epidermis
(3) The size of the vascular bundles are dependent on the size of the veins
(4) Xylem is positioned closed to the lower epidermis as compared to phloem

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**101.** Bulliform cells in the leaves are responsible for

(1) Protecting the plant from salt stress

(2) Providing large spaces for assimilation of sugars

(3) Increased photosynthesis in monocots

(4) Inward curling of leaves due to water stress

**102.** Which of the following statements about pteridophytes is **incorrect**?

(1) The spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic protonema.

(2) The sporangia may produce two kinds of spores, macro and micro, by meiosis in spore mother cells.

(3) Fusion of motile male gamete with the egg forms zygote.

(4) Zygote develops into young embryo within the female gametophyte.

**103.** The members of Chlorophyceae, Phaeophyceae and Rhodophyceae can be differentiated from each other in terms of

a. Chief photosynthetic pigments
b. Types of stored food material
c. Components of cell wall
d. Number, point of attachment and presence or absence of locomotory structures in gametes
e. Habitat such as aquatic and terrestrial

The **correct** one(s) is/are

(1) Only a, c and e

(2) Only d

(3) Only b, c and d

(4) Only b and d

**104.** On the basis of complexity of main plant body, arrange the following in ascending order.

a. *Ulothrix*
b. *Sphagnum*
c. *Marchantia*
d. *Ginkgo*
e. *Equisetum*

Select the **correct** option.

(1) d, e, c, b, a

(2) a, b, c, e, d

(3) b, a, e, c, d

(4) a, c, b, e, d

**105.** Match the following columns and select the **correct** option.

| | Column-I | | Column-II |
|---|---|---|---|
| i. | Mycorrhiza | a. | Originates from the gametophytic body in bryophytes and can be unicellular or multicellular. |
| ii. | Rhizome | b. | Represents the association between two autotrophic organisms |
| iii. | Rhizoids | c. | This organ possesses well differentiated vascular tissue |
| iv. | Coralloid roots | d. | Represents the association between autotrophic and heterotrophic organisms |

| | i | ii | iii | iv |
|---|---|---|---|---|
| (1) | d | a | c | b |
| (2) | b | a | c | d |
| (3) | d | c | a | b |
| (4) | a | b | d | c |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**106.** Agar can be obtained from

(1) *Chlorella*

(2) *Equisetum*

(3) *Fucus*

(4) *Gelidium*

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**107.** Identify the plant shown in the below given diagram and choose the statement that is **not** true for it.



(1) It belongs to a division which includes first terrestrial plants to possess vascular tissues.

(2) Its sporophyte bears sporangia that are present on sporophylls called macrophylls.

(3) It is found in cool, damp, shady places and sex organs antheridia and archegonia are produced on the main plant body, *i.e.*, sporophyte.

(4) It is frequently grown as ornamentals and has large leaves.

**108.** Archegoniophore is a

(1) Flask-shaped structure that produces egg cell

(2) Structure that is attached to the sex organ which produces motile gametes

(3) Stalk bearing multicellular flask-shaped structures

(4) Haploid structure bearing spirally arranged leaves

**109.** State **true (T)** or **false (F)** for the given statements and choose the **correct** option.

a. Heterosporous plants are evolutionarily advanced than the homosporous plants.

b. During post-fertilisation development ovaries do not develop into fruits in gymnosperms because of the degeneration of ovary wall.

c. The female gametophyte in heterosporous pteridophytes are not permanently retained on the parent sporophyte.

d. In *Cycas*, male and female cones are borne on same tree.

| | a | b | c | d |
|---|---|---|---|---|
| (1) | T | F | T | T |
| (2) | F | T | T | T |
| (3) | T | F | T | F |
| (4) | F | F | T | F |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**110.** The first group of kingdom plantae exhibiting pollination are

(1) Gymnosperms

(2) Angiosperms

(3) Pteridophytes

(4) Bryophytes

**111.** In gymnosperms the plant parts which consist of two generations – one within the other are

a. Spores inside the gametophyte

b. Male gametophyte within the microsporangia

c. Egg cell within the female sex organ in gymnosperm

d. Male gametes within germinating pollen grain

(1) b only

(2) b and c only

(3) a, b and c only

(4) c and d only

**112.** The stems are unbranched in

(1) *Cedrus*

(2) *Sequoia*

(3) *Cycas*

(4) *Ginkgo*

**113.** Which of the following is **incorrectly** matched?

(1) *Marchantia* – Dioecious gametophyte

(2) *Sphagnum* – Monoecious gametophyte

(3) *Pinus* – Monoecious sporophyte

(4) *Funaria* – Dioecious sporophyte

**114.** Read the following Assertion (A) and Reason (R) statements and select the correct option.

**Assertion (A):** The antennae molecules of light harvesting complex absorb light and gets excited, and this excitation energy is transferred to the reaction centre.

**Reason (R):** The excited state of chlorophyll molecule of reaction centre has low energy than that of the ground state of the antenna molecule.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

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**115.** Match the following columns and select the correct option.

| | Column-I | | Column-II |
|---|---|---|---|
| a. | Jan Ingenhousz | i. | Based on his studies of purple and green bacteria, inference was made that $O_2$ evolved by plants comes from $H_2O$ not from $CO_2$. |
| b. | Julius von Sachs | ii. | He showed that sunlight is essential for the plant process that somehow purifies the air fouled by burning candles. |
| c. | T.W. Engelmann | iii. | He found that the green parts in plants are where glucose is made and the glucose is stored as starch. |
| d. | Cornelius van Niel | iv. | Through his experiment the first action spectrum of photosynthesis was described. |

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| | a | b | c | d |
|---|---|---|---|---|
| (1) | iii | ii | iv | i |
| (2) | ii | iii | iv | i |
| (3) | iii | iv | ii | i |
| (4) | ii | i | iii | iv |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**116.** Paper chromatography of a photosynthetic organism led to identification of major pigments that showed blue green, yellow, yellow-orange and brown colour on chromatogram. The organism most likely is

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(1) *Chlamydomonas*

(2) *Laminaria*

(3) *Porphyra*

(4) *Ulothrix*

**117.** When a $C_3$ plant is allowed to grow in an atmosphere that has higher level of $O_2$ than the normal level, it leads to

(1) Decrease in the rate of photosynthesis

(2) Increase in their productivity

(3) Increase in the rate of glucose synthesis

(4) Decrease in the rate of photorespiration

**118.** Select the **odd** one w.r.t. $C_4$ plant.

(1) Leaves have Kranz anatomy

(2) They are adapted to dry tropical regions

(3) Primary stable product of carbon reaction is pyruvic acid

(4) Organic acids are found in cells of mesophyll and bundle sheath of leaves

**119.** Biosynthetic phase of photosynthesis

(1) Involves use of ATP and NADPH

(2) Shuts down immediately when light is absent

(3) Occurs in the lumen of thylakoid

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(4) Occurs only during the day time in all plants

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**120.** Chlorophyll *a* is considered as chief pigment associated with photosynthesis as

(1) Its absorption spectrum is superimposed on absorption spectrum of chlorophyll *b* and shows the peaks in different region of light spectrum

(2) Its action spectrum is not superimposed on absorption spectrum of photosynthesis and shows the peaks in the different regions of light spectrum

(3) Its absorption spectrum and action spectrum are identical to each other

(4) Its absorption spectrum can be superimposed on the action spectrum of photosynthesis and shows the peaks in the same regions of light spectrum

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**121.** The first step in photosynthesis is the

(1) Joining of 3-carbon molecules to form glucose

(2) Formation of ATP

(3) Synthesis of NADPH

(4) Excitation of electrons of chlorophyll by photons

**122.** Identify the **incorrect** statement regarding the events occurring in Z-scheme of light reaction of photosynthesis.

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(1) The LHC of one photosystem that is at less negative redox potential as compared to the other, has reaction centre having absorption peak of 680 nm.

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(2) As electrons move through the photosystems, protons are transported across the thylakoid membrane.

(3) Plastoquinone removes proton from the stroma while transporting electron.

(4) Plastocyanin oxidises the reduced $P_{700}$ which leads to replenishment of electron that was taken up by $NADP^+$

**123.** In what order do the following steps occurs in the Hatch and Slack pathway?

i. Transport of the $C_4$ acids to the bundle sheath cells.

ii. Fixation of $CO_2$ by the carboxylation of primary $CO_2$ acceptor.

iii. Conversion of $C_4$ acid into malate or aspartate.

iv. Decarboxylation of the $C_4$ acids within the bundle sheath cell.

v. Transport of the $C_3$ acid from bundle sheath to mesophyll cell.

vi. Carboxylation reaction catalysed by the most abundant enzyme in the world.

(1) ii → iii → i → vi → v → iv

(2) ii → iii → i → iv → vi → v

(3) i → iii → iv → vi → v → ii

(4) v → i → iii → ii → vi → iv

**124.** Read the following statements and select the correct option.

**Statement A** : The fate of the pyruvate depends on the availability of oxygen and the organism.

**Statement B** : In thick, woody stems the living cells are organised in thin layers inside and beneath the bark.

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(1) Both statement A and B are correct

(2) Both statement A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

**125.** Substrate level phosphorylation occurs during the conversion of

(1) Glucose-6-phosphate → Fructose-1, 6-bisphosphate

(2) Glyceraldehylde-3-phosphate → Dihydroxy acetone phosphate

(3) 1, 3-bisphosphoglyceric acid → Triose phosphate

(4) 3-phosphoglyceric acid → 2- phosphoglycerate

**126.** $NADH + H^+$ is produced during the conversion of

(1) Glyceraldehyde-3-phosphate → 1, 3-bisphosphoglyceric acid

(2) Succinic acid → Malic acid

(3) Succinyl-CoA → Succinic acid

(4) Phosphoenol pyruvate → Pyruvic acid

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**127.** State **true (T)** or **false (F)** for the given statements and select the correct option.

a. The term glycolysis has originated from the Greek words, *glycos* for sugar and *lysis* for splitting.

b. Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate by the activity of hexokinase.

c. There are two steps of conversions in a Krebs' cycle where $FAD^+$ is reduced to $FADH_2$

| | a | b | c |
|---|---|---|---|
| (1) | T | F | T |
| (2) | T | T | F |
| (3) | F | T | F |
| (4) | T | F | F |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**128.** In an experiment, isolated thylakoids kept previously at pH 8 were equilibrated in an acid medium at pH 4. The thylakoids are then transferred to a buffer at pH 8 that contained ADP and $P_i$. It was observed during this experiment that ATP synthesis occured even in the dark. Which of the following best explains the reason for ATP synthesis in this experiment?

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(1) Transfer of thylakoids from low to high $H^+$ concentration creates proton gradient across the thylakoid membrane that drives ATP synthesis.

(2) ATP synthesis occurred because $H_2O$ in acidic medium acted as an electron donor for photophosphorylation.

(3) When the thylakoids were placed in pH 4 medium, the $H^+$ concentration in their lumen increased and again shifting to the high pH caused the breakdown of proton gradient leading to ATP synthesis.

(4) ADP and $P_i$ acted as substrates that directly phosphorylated each other in alkaline medium.

**129.** When electrons flow in the ETS of mitochondria, it results in the change in

(1) Electrochemical gradient which causes increase in proton concentration in the matrix and decrease in inter-membrane space

(2) pH of the matrix that becomes high in comparison to the inter-membrane space

(3) The rate of ADP synthesis

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(4) Total amount of $NAD^+$ oxidised in Krebs' cycle

**130.** Ubiquinone acts as electron carrier between

(1) Complex I and complex II

(2) Complex II and complex IV

(3) Complex I and complex III

(4) Complex III and complex IV

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131. In the citric acid cycle, how many steps involve decarboxylation?

(1) One

(2) Two

(3) Three

(4) Four

132. Assume that five glucose molecules provided to each isolated cell (1) and cell (2). In cell (1) mitochondria had nonfunctional complex II while cell (2) had no abnormality. Oxaloacetic acid to the matrix of mitochondria of each cell were supplied. What will be the ratio between net gain of ATP by cell (1) and cell (2)?

(1) 19 : 17

(2) 7 : 12

(3) 4 : 3

(4) 14 : 19

133. Which of the following biomolecules is common to respiration mediated breakdown of fatty acids and amino acids?

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(1) Glyceraldehyde 3-phosphate

(2) Pyruvic acid

(3) Fructose-6-phosphate

(4) Acetyl CoA

134. Read the following assertion (A) and reason (R) statements and select the correct option.

**Assertion (A)** : Presence of $O_2$ is essential for the functionality of Krebs' cycle even $O_2$ itself is not utilised in its reaction.

**Reason (R)** : Oxygen acts as the final hydrogen acceptor in ETS.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

135. Read the following statements (A to D).

A. Alcoholic fermentation is commonly seen in yeast.

B. During oxidative decarboxylation, pyruvic acid is converted into acetyl CoA by the catalytic activity of pyruvate dehydrogenase.

C. ATP synthase (complex V) consist of two major components, one of which is peripheral membrane protein complex that contains site for ATP synthesis.

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D. Krebs' cycle starts with the condensation of acetyl CoA with a 3-carbon compound to yield citric acid.

How many of the above statements is/are **correct**?

(1) Two

(2) Three

(3) Four

(4) One

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## ZOOLOGY

136. Which of the following muscles is/are striated and myogenic?

(a) Skeletal
(b) Cardiac
(c) Visceral

Select the **correct** option.

(1) a, b

(2) Only b

(3) b, c

(4) Only a

137. During relaxation of a skeletal muscle fibre

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(1) ATP hydrolysis occurs

(2) $Ca^{++}$ concentration in sarcoplasm increases

(3) $Ca^{++}$ is actively pumped back into the sarcoplasmic reticulum

(4) Myosin heads remain attached to the actin

138. A sarcomere does not contain

(1) Two Z-lines

(2) Two A-bands

(3) Two half I-bands

(4) One A-band

139. Select the **correct** option w.r.t structure of a meromyosin.

(1) HMM = Head + Cross arm

(2) LMM = Short arm + Tail

(3) HMM = Cross arm

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(4) LMM = Head

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140. Which of the following is not a function of the human skeleton?

(1) Protection of internal organs

(2) Production of blood cells

(3) Providing attachment site for skeleton muscle

(4) Regulation of body temperature

**141.** The appendicular skeleton of humans does not include

(1) Tibia and fibula

(2) Femur and humerus

(3) Floating ribs and sternum

(4) Scapula and clavicle

**142.** Consider the following w.r.t. action potential in neurons.

(a) Depolarisation involves influx of $Na^+$

(b) Repolarization involves efflux of $K^+$

(c) $Na^+ - K^+$ pump actively restores ionic gradient across resting axonal membrane

Which of the above are correct?

(1) a, b, c

(2) Only b, c

(3) Only a, c

(4) Only a, b

**143.** If ATP is suddenly removed from a contracting skeletal muscle fibre, which immediate change will be seen?

(1) Detachment of cross-bridge

(2) No breakage of cross-bridge

(3) Conformational change of troponin-tropomyosin complex

(4) Return of $Ca^{++}$ to the sarcoplasmic reticulum

**144.** At rest, which of the following is not associated with actin filaments in a sarcomere present in biceps?

(1) A-band

(2) I-band

(3) Z-line

(4) M-line

**145.** The type of joint present between atlas and axis vertebrae is

(1) Cartilaginous joint

(2) Pivot joint

(3) Saddle joint

(4) Hinge joint

**146.** Which of the following pairs is **incorrectly** matched w.r.t humans?

(1) Hinge joint – Knee joint, elbow joint

(2) Ball and socket joint – Hip joint, shoulder joint

(3) Fibrous joint – Pubic symphysis

(4) Saddle joint – Carpal and metacarpal of thumb

**147.** Select the option representing the **correct** sequence of events in a skeletal muscle contraction.

(a) Depolarization of sarcolemma

(b) Cross-bridge formation

(c) $Ca^{++}$ release from SR

(d) ATP hydrolysis

(e) Sliding of filaments

(1) a → c → d → b → e

(2) a → b → c → e → d

(3) a → c → b → d → e

(4) c → a → d → b → e

**148.** In an adult human, the total number of lumbar vertebrae is equal to the number of

(1) Carpals in one forelimb

(2) Metatarsals in one hindlimb

(3) Floating ribs

(4) Cranial bones

**149.** In a skeletal muscle, after arrival of neural impulse, removal of extracellular $Ca^{2+}$ does not immediately stop contraction because

(1) $Ca^{2+}$ entry through sarcolemma is required only for depolarisation

(2) Contraction depends mainly on $Ca^{2+}$ released from sarcoplasmic reticulum

(3) Extracellular $Ca^{2+}$ is stored inside mitochondria

(4) $Ca^{2+}$ is replaced by $Na^+$ present inside the sarcomere

**150.** Flexibility of the thoracic cage during breathing is an outcome of

(1) Elastic recoil of lungs and presence of synovial joints between ribs and cervical vertebrae

(2) Presence of fibrous joints between ribs and sternum

(3) Elasticity of hyaline cartilages between ribs and sternum and slight movement at joints between ribs and vertebrae

(4) Specific rib movements and monocephalic ribs

**151.** A newborn human baby has 33 separate vertebrae while an adult human has 26. Which of the following cause the reduction in number?

(1) Fusion of lumbar vertebrae and sacral vertebrae respectively

(2) Fusion of thoracic and lumbar vertebrae

(3) Fusion of cervical and thoracic vertebrae

(4) Fusion of sacral vertebrae and coccygeal vertebrae respectively

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**152.** Synovial fluid lubricates joints and usually does not
(a) Supply nutrients to articular cartilage
(b) Increase friction for mobility
(c) Primarily store $Ca^{2+}$ ions
(d) Strengthen ligaments which connect bones to muscles
Select the **correct** option.

(1) a, b, c

(2) a, c, d

(3) b, c, d

(4) a, b, d

**153.** The transmission of nerve impulses across the most abundantly present synapses in humans occur by

(1) Electrical transmission

(2) Movement of $Na^+$ through gap junctions present between pre and post-synaptic neurons

(3) Neurotransmitters released from pre-synaptic neurons.

(4) Direct contact of axon and dendrite

**154.** The autonomic nervous system regulates

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(1) Voluntary muscle movements

(2) Only reflex actions

(3) Critical thinking and memory directly

(4) Activities of internal visceral organs

**155.** Injury to medulla oblongata can lead to death because

(1) It possesses centres for control of body temperature and hunger

(2) It possesses centres for control of breathing and cardiovascular reflexes

(3) It controls memory and possesses centres for thirst

(4) It interconnects different regions of the cerebral cortex

**156.** Which of the following is not the function of hypothalamus?

(1) Thermoregulation

(2) Regulation of endocrine activities

(3) Coordination of voluntary movements

(4) Control of hunger

**157.** If a neuron with normal intracellular concentration of $Na^+$ is kept in an external medium containing high concentration of $K^+$ and $Na^+$ compared to its axoplasm, then which change will most likely occur in the neuron?

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(1) $Na^+/K^+$ pump becomes non-functional

(2) Influx of $K^+$ will occur

(3) Probability of action potential generation decreases

(4) Neuron becomes repolarised

**158.** Read the following statements
**Statement A:** Like the cerebrum, the cerebellum has a highly folded surface that greatly increases the surface area of its outer gray matter, allowing for a greater number of neurons.
**Statement B:** The primary function of the cerebellum is to evaluate how well movements initiated by motor areas in the cerebrum are actually being carried out.
Choose the **correct** option.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

**159.** **Assertion (A):** Chemical synapses exhibit principle of one-way transmission that makes them highly desirable for transmitting neural signals.
**Reason (R):** A one-way transmission allows signals to be directed toward specific goals so that nervous system can perform myriad of functions like sensation, motor control, memory *etc.*
In the light of above statements, choose the **correct** option.

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(1) Both (A) and (R) are true and (R) explains (A) correctly

(2) Both (A) and (R) are true but (R) does not explain (A) correctly

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

**160.** When a person closes his eyes and tries to touch the tip of his nose with his finger, which part of the brain ensures the accurate movement and coordination?

(1) Pons

(2) Hypothalamus

(3) Medulla oblongata

(4) Cerebellum

**161.** During fight or flight response, there is sympathetic activation that leads to
(a) Dilation of pupil
(b) Decrease in rate of respiration
(c) Increase in heart beat rate
(d) Decrease in blood glucose level
Select the **correct** option.

(1) a, c

(2) a, b

(3) c, d

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(4) b, d

**162.** If the thyroid gland of a person is surgically removed but the pituitary and hypothalamus remain intact, then which of the following changes will be observed after a few days?

(1) TRH↑, TSH↑, T4↓

(2) TRH↓, TSH↑, T4↑

(3) TRH↑, TSH↓, T4↓

(4) TRH↓, TSH↓, T4↑

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**163.** Which of the following statements best explains why the brain stem, despite its small size, is associated with a wide range of essential functions?

(1) It evolved last during the human evolution and therefore has the complex function

(2) It contains all the centres for learning and thinking

(3) It lies between the forebrain and spinal cord, so all neural pathways must pass through it

(4) It has a larger number of neurons than cerebrum

**164.** Which of the following ions acts as the actual trigger for cross-bridge formation?

(1) $Ca^{++}$

(2) $Cl^-$

(3) $Na^+$

(4) $K^+$

**165.** Match column I, II and III w.r.t. structural components of a neuron, their functions and physiological significance.

| | Column I | | Column II | | Column III |
|---|---|---|---|---|---|
| (a) | Dendrites | (p) | Increases the impulse conduction velocity | (i) | High density of voltage gated $Na^+$ channels |
| (b) | Axon hillock | (q) | Transmits signal to next neuron | (ii) | Contains Nissl's bodies |
| (c) | Axon terminal | (r) | Site of initiation of action potential | (iii) | Interrupted at nodes of Ranvier |
| (d) | Myelin sheath | (s) | Receive impulses and conduct them toward cell body | (iv) | Contains synaptic vesicles |

Choose the **correct** option.

(1) (a) –(s), (ii); (b) – (r), (i); (c) – (q), (iv); (d) – (p), (iii)

(2) (a) –(p), (iv); (b) – (q), (iii); (c) – (r), (ii); (d) – (s), (i)

(3) (a) –(r), (iii); (b) – (q), (ii); (c) – (p), (i); (d) – (s), (iv)

(4) (a) –(q), (i); (b) – (p), (iv); (c) – (s), (iii); (d) – (r), (ii)

**166.** Which of the following hormones is not under the direct control of hypothalamic releasing hormones?

(1) TSH

(2) GH

(3) FSH

(4) Insulin

**167.** An increase in growth hormone levels in adults especially in middle age leads to

(1) Addison's disease

(2) Graves' disease

(3) Acromegaly

(4) Gigantism

**168.** A patient has low sodium and high potassium levels in his blood. Which hormone is most likely deficient in him?

(1) Aldosterone

(2) Oxytocin

(3) ADH

(4) PTH

**169.** If the posterior pituitary is damaged, release of which hormone will be affected and what physiological change will it cause?

(1) GH – Decreases growth

(2) ADH – Increased urine output

(3) ACTH – Low cortisol secretion

(4) Oxytocin – Increased blood volume

**170.** If plasma calcium level in a healthy man decreases below normal levels, then which of the following series of events will occur?

(1) Calcitonin↑ → bone deposition ↑ → plasma $Ca^{++}$↑

(2) PTH↑ → bone resorption↑ → Plasma $Ca^{++}$↑

(3) PTH↓ → $Ca^{++}$ absorption↓ → Plasma $Ca^{++}$↑

(4) Calcitonin ↑ → $Ca^{2+}$ renal reabsorption ↑ → Plasma $Ca^{++}$↑

**171.** Which of the following hormones directly controls the water permeability of the collecting duct present in kidneys?

(1) Renin

(2) Cortisol

(3) ADH

(4) ACTH

**172.** A 45 year old man has a mass in the sella turcica that compresses the portal vessels, disrupting pituitary access to hypothalamic secretions. The secretion rate of which of the following hormones would most likely not be affected in this patient?

(1) Adrenocorticotropic hormone

(2) Gonadotropins

(3) Thyroid stimulating hormone

(4) Oxytocin

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**173.** Consider the following statements:
(a) It Consists of fibre tract that interconnect different regions of the brain
(b) It contains pneumotaxic centre
(c) It is a part of brain stem
(d) It is directly involved in regulation of emotional reactions such as motivation
Which of the above statements are true w.r.t pons?

(1) a, b, c

(2) b, c, d

(3) a, c, d

(4) a, b, d

**174.** **Assertion (A):** Epinephrine acts both as a hormone and as a neurotransmitter.
**Reason (R):** Epinephrine is secreted by adrenal cortex and is also released at chemical synapses by sympathetic neurons.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) explains (A) correctly

(2) Both (A) and (R) are true but (R) does not explain (A) correctly

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

**175.** Consider the following statements:
(a) Thyroid hormones and adrenal cortical hormones are involved in maintenance of water and electrolyte balance.
(b) Renin from kidneys, thyroxine from thymus and cortisol from adrenal are involved in erythropoiesis
(c) Carbohydrate metabolism can be regulated by thyroxine and glucocorticoids.
(d) CCK acts on target cells in gall bladder through second messenger.
Which of the above statements are not incorrect?

(1) a, b, c

(2) a, c, d

(3) b, c, d

(4) a, b, d

**176.** Listed below are some hormones
(a) ANF (b) Thymosin
(c) Thyroxine (d) Progesterone
(e) Androgenic steroids
Which among the above is/are non-steroidal hormones that interact with intracellular receptors?

(1) b, c

(2) d, e

(3) Only a

(4) Only c

**177.** Select the option that correctly represents the mechanism of action of FSH.

(1) FSH → Interacts with membrane-bound receptor → Generation of $2^{nd}$ messenger i.e, ATP → Physiological responses → Genome responses

(2) FSH → Binds with intracellular receptor → Generation of $2^{nd}$ messenger → Biochemical responses → Physiological responses

(3) FSH → Binds with membrane-bound receptors → Generation of $2^{nd}$ messenger → Biochemical responses → Physiological responses

(4) FSH → Binds with intracellular receptor → Acts on genome → mRNA → Protein → Physiological responses

**178.** Observe the combinations w.r.t. hormone, its source and its effect given below and select the correct match.

| | Source | Hormone | Effect |
|---|---|---|---|
| 1. | Adrenal medulla | Epinephrine | Lipogenesis |
| 2. | Adrenal cortex | Cortisol | Increases blood pressure |
| 3. | Pancreas | Glucagon | Glycogenesis |
| 4. | Leydig cells | Luteinizing hormone | Regulates the development and maturation of epididymis |

(1) 1

(2) 2

(3) 3

(4) 4

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**179.** Match column I and II w.r.t. hormones and their respective source.

| | Column I | | Column II |
|---|---|---|---|
| (a) | Tetraiodothyronine | (i) | Testis |
| (b) | Testosterone | (ii) | Pancreas |
| (c) | Insulin | (iii) | Adrenal |
| (d) | Cortisol | (iv) | Thyroid |

Choose the **correct** match.

(1) (a) – (iv) ; (b) – (i); (c) – (ii); (d) – (iii)

(2) (a) – (iv); (b) – (ii); (c) – (iii); (d) – (i)

(3) (a) – (i); (b) – (iv); (c) – (ii); (d) – (iii)

(4) (a) – (iv); (b) – (iii); (c) – (i); (d) – (ii)

**180.** Read the statements A, B and C carefully and select the **correct** option.

**Statement (A):** Increase in blood levels of estrogen before ovulation stimulates ovarian follicles to release LH and cause LH surge.

**Statement (B):** Role of glucagon and insulin w.r.t regulation of glucose homeostasis is a kind of antagonistic mechanism.

**Statement (C):** Diabetes mellitus is always caused due to hyposecretion of glucagon and hypersecretion of insulin.

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(1) Statements (A) and (B) are correct but statement (C) is incorrect

(2) Statements (B) and (C) are incorrect but statement (A) is correct

(3) Statement (A) and (C) are incorrect but statement (B) is correct

(4) Statements (A), (B) and (C) are incorrect

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